

KATHMANDU UNIVERSITY

End Semester Examination

August/September, 2017

Level : B. E./B.Sc./B. Tech.

Year : I

Course : MATH 104

Semester : II

Exam Roll No. :

Time: 30 mins.

F. M. : 20

Registration No.:

Date :

SECTION "A"

[10Q × 1 = 10 marks]

Fill in the blank space(s) by most appropriate word(s) or symbol(s).

- Equation of circle passing through the origin centered on negative Y-axis with radius a is _____
- The equation in cylindrical coordinate system of the equation in spherical coordinate system $\rho = 5\cos\phi$ is _____
- If $\vec{u}$ is a differentiable vector function of t of constant length, then $\vec{u} \cdot \frac{d\vec{u}}{dt} =$ _____
- The curvature of a circle of radius a is _____
- A region in the plane is bounded if _____
- The gradient of a differential function at a point is always _____
- If $\vec{F}$ is conservative then the work done around any closed loop equals _____
- The area of the surface $f(x, y, z) = c$ over a closed and bounded plane region R is _____
- The Fourier coefficient a_n of the periodic function $f(t)$ of period T is given by the Euler's formula _____
- $\int_1^e \int_1^e \int_1^e \frac{1}{xyz} dx dy dz =$ _____

SECTION "B"

[10 Q. × 1 = 10 marks]

Fill in the blank space(s), DO NOT TICK, by choosing the most appropriate answer from among the given ones.

- The polar coordinate for Cartesian coordinate $(\sqrt{3}, 1)$ is _____
 $[(-2, -5\pi/6); \quad (-2, -7\pi/6); \quad (2, -5\pi/6); \quad (2, -7\pi/6)]$
- The equation $\frac{8}{2-4\sin\theta}$ represents _____
[Circle ; Ellipse; Hyperbola; Parabola]

13. If $f(x, y) = y \sin xy$ then $\frac{\partial f}{\partial y} =$ _____
 $[x^2 \cos xy + \sin xy; \quad xy \cos xy + \sin xy;$
 $y^2 \cos xy + \sin xy; \quad xy \cos xy]$
14. Unit tangent vector of a curve $\vec{r}(t) = \cos t \vec{i} + \sin t \vec{j}$ is _____
 $[\vec{v}; \quad \sin t \vec{i} - \cos t \vec{j}; \quad -\sin t \vec{i} - \cos t \vec{j}; \quad \cos t \vec{i} + \sin t \vec{j}]$
15. The integral $\iint y \delta(x, y) dA$, where δ is the density represents _____
 $[\text{Mass}; \quad \text{Area}; \quad \text{Moment}; \quad \text{Volume}]$
16. The work done by the conservative field $\vec{F} = yz\vec{i} + xz\vec{j} + xy\vec{k}$ along any smooth curve C joining the point $(-1, 3, 9)$ to $(1, 6, -4)$ is _____
 $[3; \quad 4; \quad -3; \quad -4]$
17. The divergence of $\vec{F} = 2xz\vec{i} - xy\vec{j} - \vec{k}$ is given by the expression _____
 $[2z + x - 1; \quad 2z - x + 1; \quad 2z - x - 1; \quad 2z + x + 1]$
18. $\int_0^\infty e^{-x^2} dx =$ _____
 $[\frac{\sqrt{\pi}}{2}; \quad \frac{\pi}{2}; \quad \frac{1}{\sqrt{\pi}}; \quad \frac{2}{\sqrt{\pi}}]$
19. The value of $\beta\left(\frac{1}{3}, \frac{2}{3}\right) =$ _____ -
 $[\frac{\sqrt{3}}{2}\pi; \quad \frac{2}{\sqrt{3}}\pi; \quad \frac{3}{\sqrt{2}}\pi; \quad \frac{\sqrt{2}}{3}\pi]$
20. _____ is the primitive period of $f(x) = \sin 4x$.
 $[\frac{\pi}{2}; \quad \pi; \quad 2\pi; \quad 4\pi]$

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Level : B.E./B.Sc./B. Tech.

Year : I

Time : 2hrs. 30 mins.

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Semester : II

F. M. : 55

SECTION "C"

[4 Q. $\times$ 7 = 28 marks]

1. Graph the lemniscate $r^2 = 6 \cos 2\theta$. Find the area inside the lemniscates $r^2 = 6 \cos 2\theta$ and outside the circle $r = \sqrt{3}$. [3+4]
2. Define directional derivative and state its properties. Using this definition, find the derivative of $f(x, y) = x^2 + xy$ at $P_0(1, 2)$ in the direction of unit vector $\vec{u} = \frac{1}{\sqrt{2}}\vec{i} + \frac{1}{\sqrt{2}}\vec{j}$. Find the directions in which $f(x, y) = \frac{x^2}{2} + \frac{y^2}{2}$ increases and decreases most rapidly. [1+1+3+2]
3. State Fubini's theorem (stronger form). Find the volume of the solid whose base is the region in the xy – plane that is bounded by the parabola $y = 4 - x^2$ and the line $y = 3x$, while the top of the solid is bounded by the plane $z = x + 4$. [2+5]

OR

State Fubini's theorem (stronger form). Sketch the region of integration and then evaluate the integral $\int_0^{2\sqrt{\ln 3}} \int_{y/2}^{\sqrt{\ln 3}} e^{x^2} dx dy$. [2+5]

4. State Stoke's theorem. Find the circulation of the field $\vec{F} = (x^2 - y)\vec{i} + 4z\vec{j} + x^2\vec{k}$ around the curve C in which the plane $z = 2$ meets the cone $z = \sqrt{x^2 + y^2}$, counterclockwise as viewed from above. [2+5]

SECTION "D"

[9 Q. $\times$ 3 = 27 marks]

5. Show that the function $f(x, y) = \frac{x^4}{x^4 + y^2}$ have no limit as $(x, y) \rightarrow (0, 0)$.
6. Define Gamma function. Prove that $\Gamma(n + 1) = n\Gamma(n)$.

OR

Prove that $B(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$

7. Find the length of spiral $r = e^\theta / \sqrt{2}$ from $\theta = 0$ to $\theta = \pi$.
8. Solve initial value problem: $\frac{d\vec{r}}{dt} = (t^3 + 4t)\vec{i} + t\vec{j} + 2t^2\vec{k}$, $\vec{r}(0) = \vec{i} + \vec{j}$.
9. Find $\vec{T}$, $\vec{N}$ and κ for the curve $\vec{r}(t) = \frac{t^3}{3}\vec{i} + \frac{t^2}{2}\vec{j}$, $t > 0$ where the symbols have their usual meanings.
10. Find the points on the curve $xy^2 = 54$ nearest the origin.

11. Evaluate: $\int_{-1}^1 \int_0^{2\pi} \int_0^{1+\cos\theta} 4r \, dr d\theta dz$.
12. Calculate the outward flux for the field $\vec{F} = (x^2 + 4y)\vec{i} + (x + y^2)\vec{j}$ across the square bounded by the lines $x = 0, x = 1, y = 0, y = 1$.

OR

Evaluate $\int_C \vec{F} \cdot \vec{T} \, ds$ for the vector field $\vec{F} = x^2\vec{i} - y\vec{j}$ along the curve $x = y^2$ from (4,2) to (1,-1).

13. Find the Fourier series of the function $f(x) = \begin{cases} 0 & \text{if } -\pi < x < 0 \\ x & \text{if } 0 < x < \pi \end{cases}$.